

Matthew Au-Yeung

 mattheway.com |  [M4TTH3](https://github.com/M4TTH3) |  [matthewauyeung](https://www.linkedin.com/in/matthewauyeung) |  mw2auyeu@uwaterloo.ca |  226-751-5487

SKILLS

Languages: C#, Kotlin, Java, Python, C/C++, JavaScript/TypeScript, HTML/CSS, SQL, Bash
Frameworks: ASP.NET Core, Spring Boot, Android, React Native, Jetpack Compose, React, Node.js, FastAPI
Databases/Cloud: PostgreSQL, MSSQL Server, MongoDB, Azure, Dynamics365, Docker, Nginx

EDUCATION

University of Waterloo

Waterloo, ON

Candidate for Bachelors of Computer Science, Honours Co-op

2022 – 2026

- GPA: 3.66
- Courses: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Intro to Compilers

EXPERIENCE

Incoming Software Developer

Jan 2025 — Present

Ford Motor Company

Waterloo, ON

- Developing mobile integration features for Ford's Infotainment system using Kotlin

Compilers Research Assistant

May 2024 – Jan 2025

University of Waterloo

Waterloo, ON

- Developed a Kotlin-based **MIPS assembly runtime environment** used annually by **over 1,000 students** for the Compilers Foundation course
- Engineered a **graphical command-line MIPS debugger**  in Kotlin that supports stepping, breakpoints, I/O, and processor state visuals, reducing course Piazza queries by **60%**
- Built a compiler for an educational language implementing **20%** of C's functionality, targeting MIPS assembly
- Created an ANTLR grammar for TopFormFlat file minimization (a bracket-nesting newline limit) by abstracting common CFG rules, expanding support from only C to **9 languages**

Software Developer

May 2024 – Aug 2024

Township of Centre Wellington

Fergus, ON

- Shipped **10 full-stack MVC pages** to manage and interact with **over 1,000,000 rows** of government data, using C# ASP.NET Core, Entity Framework, SQL Server, and JQuery
- Implemented policy-based authorization with controller middleware to enforce access across **28 municipalities**
- Improved communication for **100+ officials** by developing real-time notes using SignalR Websockets and Quill.js
- Integrated Github Actions for automated deployment to **Azure App Service**; improved build speed by **50%** via Webpack-based TypeScript transpilation and bundling

Data Science & Administrative Assistant

May 2023 – Aug 2023

Bio-Ag Consultants and Distributors

Wellesley, ON

- Implemented Python scripts to streamline access to company metrics, sales, and inventory from Dynamics365, saving **over 3 hours daily** of manual data logging
- Analyzed inventory datasets of over **10,000 items** to uncover **\$30,000** in losses due to manufacturing errors

PROJECTS

Timestamp | *Android, Kotlin, Jetpack Compose, Spring Boot, PostgreSQL, Firebase Auth, Docker* Sept 2024

- Led a **team of 4** to develop a privacy-focused Android meetup app that obscures locations until arrival, and alternatively provides arrival estimates
- Developed a Spring Boot backend implementing CRUD operations with Hibernate JPA and **PostgreSQL**; and integrated Graphhopper's routing engine to reduce routing requests from **500ms to 10ms**
- Built a modern UI using **Material3** design, streamlining data flow with an MVVM architecture reusing Singleton Ktor repositories and activity-scoped ViewModels
- Deployed an NGINX reverse proxy for self-hosted PostgreSQL, Spring Boot, and Photon Geocoder containers, managing horizontal scaling and load-balancing across **3 server instances**

Chess | *C++, React, Node.js, Express, Socket.io, Docker*

Nov 2023

- Launched a fully-featured chess game in **C++**, incorporating multiple design patterns and MVC to achieve high efficiency and extensibility
- Combined an interactive **React** interface with a **Socket.io** server, enabling real-time interaction with the chess game